

SUPPORTING INFORMATION

Internal Camera-based Reconstruction and Closed-loop Control of Soft Robotic Arms

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1 | Notation and Terminology

The symbols and the corresponding explanations used throughout the paper are reported in Table S1 and Table S2.

2 | PCC and PPC Model Details

2.1 | PCC Model Derivation

Under the PCC hypothesis, each arc included in a segment has the same curvature, and all arcs bend on parallel planes. The four boundary arcs placed at distance d from the central axis along the x - and y -directions have lengths [1]:

$$L_1 = L - \theta d \cos \phi \quad (\text{S1a})$$

$$L_2 = L + \theta d \cos \phi \quad (\text{S1b})$$

$$L_3 = L - \theta d \sin \phi \quad (\text{S1c})$$

$$L_4 = L + \theta d \sin \phi \quad (\text{S1d})$$

Taking pairwise differences yields the q -parameters:

$$\Delta_x = \frac{L_2 - L_1}{2} = \theta d \cos \phi \quad (\text{S2a})$$

$$\Delta_y = \frac{L_4 - L_3}{2} = \theta d \sin \phi \quad (\text{S2b})$$

where L_1, L_2, L_3, L_4 are the arc lengths of the cylinder soft segment.

2.2 | PPC Bending and Coefficients Initialization

Let $T^s \in SE(3)$ denote the homogeneous transformation from the base frame to any point along the section axis:

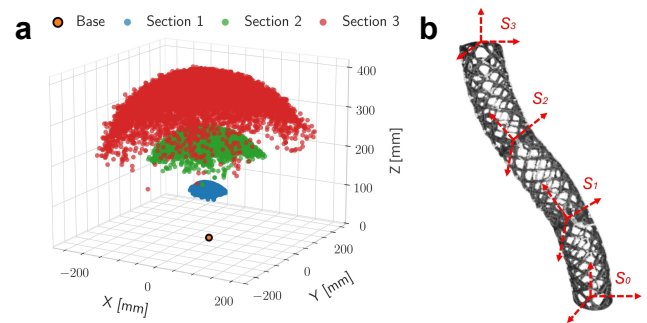


FIGURE S1 | (a) Workspace of the multi-segment robot. Sampled positions of the first, second, and third ends are depicted as blue, green, and red dots. (b) Main reference systems during modeling and data acquisition.

$$T^s = \begin{bmatrix} R^s & t^s \\ 0 & 1 \end{bmatrix} \quad (\text{S3})$$

The motion capture system is used to recover the measured rotation matrices $\hat{R}^s \in \mathbb{R}^{3 \times 3}$ at the subsections of each module. The ground-truth bending direction and angle estimates $\hat{\phi}_s, \hat{\theta}_s$ can be estimated as:

$$\hat{\phi}_s = \arctan2(R_{32}^s, R_{31}^s) \quad (\text{S4a})$$

$$\hat{\theta}_s = \arccos R_{33}^s \quad (\text{S4b})$$

Polynomial coefficients are initialized analytically by substituting $s = \frac{1}{3}, \frac{2}{3}, 1$ into $\theta(s) = c_0 s + \frac{1}{2} c_1 s^2 + \frac{1}{3} c_2 s^3$ and solving the corresponding linear systems according to the optimization algorithm described in [2].

2.2.1 | Constant Model ($c_1 = c_2 = 0$)

Only the tip orientation is used:

$$\hat{c}_0 = \frac{\hat{\theta}_1}{L} \quad (S5)$$

2.2.2 | Affine model ($c_2 = 0$)

The first and third subsections yield:

$$\begin{cases} \hat{c}_0 \left(\frac{L}{3}\right) + \frac{1}{2}\hat{c}_1 \left(\frac{L}{3}\right)^2 = \hat{\theta}_{\frac{1}{3}} \\ \hat{c}_0 L + \frac{1}{2}\hat{c}_1 L^2 = \hat{\theta}_1 \end{cases} \quad (S6)$$

with closed-form solution:

$$\hat{c}_0 = \frac{9\hat{\theta}_{\frac{1}{3}} - \hat{\theta}_1}{2L} \quad (S7a)$$

$$\hat{c}_1 = \frac{3\hat{\theta}_1 - 9\hat{\theta}_{\frac{1}{3}}}{L^2} \quad (S7b)$$

2.2.3 | Quadratic model

All three subsections yield:

$$\begin{cases} \hat{c}_0 \left(\frac{L}{3}\right) + \frac{1}{2}\hat{c}_1 \left(\frac{L}{3}\right)^2 + \frac{1}{3}\hat{c}_2 \left(\frac{L}{3}\right)^3 = \hat{\theta}_{\frac{1}{3}} \\ \hat{c}_0 \left(\frac{2L}{3}\right) + \frac{1}{2}\hat{c}_1 \left(\frac{2L}{3}\right)^2 + \frac{1}{3}\hat{c}_2 \left(\frac{2L}{3}\right)^3 = \hat{\theta}_{\frac{2}{3}} \\ \hat{c}_0 L + \frac{1}{2}\hat{c}_1 L^2 + \frac{1}{3}\hat{c}_2 L^3 = \hat{\theta}_1 \end{cases} \quad (S8)$$

with closed-form solution:

$$\hat{c}_0 = \frac{2\hat{\theta}_1 - 9\hat{\theta}_{\frac{2}{3}} + 18\hat{\theta}_{\frac{1}{3}}}{2L} \quad (S9a)$$

$$\hat{c}_1 = \frac{-9\hat{\theta}_1 + 36\hat{\theta}_{\frac{2}{3}} - 45\hat{\theta}_{\frac{1}{3}}}{L^2} \quad (S9b)$$

$$\hat{c}_2 = \frac{27\hat{\theta}_1 - 81\hat{\theta}_{\frac{2}{3}} + 81\hat{\theta}_{\frac{1}{3}}}{2L^3} \quad (S9c)$$

Starting from these guesses, the coefficients are refined by the Frobenius-norm optimization described in the main text.

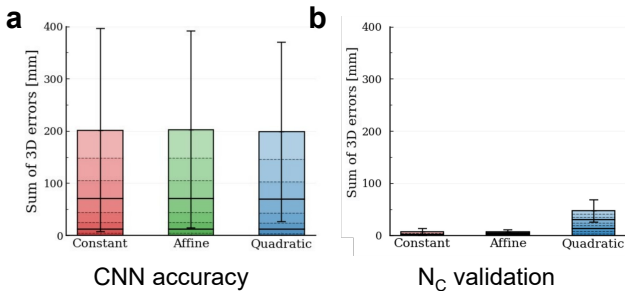


FIGURE S2 | (a) 3D reconstruction accuracy of the curvature models at reference points placed every ninth of the manipulator length. (b) Errors induced by the N_C approximation with respect to the motion-capture ground-truth.

3 | Deep Learning Architecture Details

3.1 | CNN Architecture and Training Parameters

Regarding the Convolutional Neural Network (CNN) feature extractors, the networks f_C^1 and f_C^3 are trained independently for the three kinematic models (quadratic, affine, and constant), with their outputs defined by the corresponding curvature parameters. For network initialization across the multi-section arm, f_p^3 was initialized from f_p^1 by modifying the first convolutional layer, and the resulting backbone was subsequently used to initialize the curvature estimators f_C^3 .

During training, outputs were normalized for stability, with all angular quantities reparameterized using sine and cosine. The regressors were trained using the Huber loss ($\mathcal{L}$), defined as [3]:

$$\mathcal{L}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta, \\ \delta|y - \hat{y}| - \frac{1}{2}\delta^2 & \text{otherwise,} \end{cases} \quad (S10)$$

where y represents the ground-truth value, $\hat{y}$ is the predicted value, and δ is the threshold parameter that dictates the transition from quadratic to linear penalization.

3.2 | Spatial Attention Formulation

To visualize the image regions most influencing the CNN predictions, we compute attention regions by backpropagating the Huber loss $\mathcal{L}$ to the last convolutional layer. Let the feature map A have K channels $A^k \in \mathbb{R}^{u \times v}$. Denoting the prediction as $\hat{y}$, the importance weight of channel k is calculated as:

$$\alpha_k = \frac{1}{uv} \sum_{i=1}^u \sum_{j=1}^v \left(-\frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial A_{ij}^k} \right) \quad (S11)$$

The final activation heatmap, which highlights the regions that positively contribute to reducing the loss, is then evaluated as [4]:

$$L_{\text{Grad-CAM}} = \text{ReLU} \left(\sum_{i=1}^K \alpha_k A^k \right) \quad (S12)$$

Similar spatial attention results are observed across all the single-section networks, showing that the network relies predominantly on structural deformations along the bending direction. However, this visual analysis was not performed for the multi-section case since the individual visual contributions of the different segments are difficult to disentangle.

4 | Data Acquisition Details

4.1 | Dataset Composition

The composition of the datasets for the single- and multi-segment manipulators is shown in Table S3 and Table S4, respectively. The sampled workspace of the multi-segment robot is shown in Fig. S1a, while the main reference systems of interest in Fig. S1b.

4.2 | Multi-Section Data Acquisition

The wide range of motion of the multi-segment manipulator causes severe occlusions that prevent simultaneous tracking of

TABLE S1 | Notation and terminology: geometry, kinematics, and actuation.

Symbol	Description
<i>Geometry and Kinematics</i>	
d	Radius of the module cross-section.
s	Normalized arc length along a section axis ($s \in [0, 1]$).
L	Central axis length of a module.
L_i	Arc length of the i -th outer boundary segment ($i \in \{1, 2, 3, 4\}$).
α	Kinematic parameterization vector comprising bending angle, direction, and length (θ, ϕ, L) .
$\theta(s)$	Local bending angle along the central axis at normalized arc length s .
ϕ	Bending direction of the module.
q	Kinematic parameterization vector based on base arc length differences $[\Delta_x, \Delta_y, L]^T$.
Δ_x, Δ_y	Base arc length differences along the local x - and y -axes.
Δ	Magnitude of the base arc length deformation ($\sqrt{\Delta_x^2 + \Delta_y^2}$).
$c(s)$	Polynomial curvature function at normalized arc length s .
c_0, c_1, c_2	Coefficients of the polynomial curvature approximation.
C	Vector containing the full-body curvature parameters $([c_0, c_1, c_2, L, \phi]^T)$.
$\hat{\theta}_s, \hat{\phi}_s$	Local bending angle and direction estimated from ground-truth motion capture data.
<i>State and Actuation</i>	
p	Full tip pose vector $p = [t, r]^T$, defined with respect to the base reference frame.
t	Translation/position vector $[x, y, z]^T$.
r	Euler orientation vector $(\psi_{\text{yaw}}, \theta_{\text{pitch}}, \phi_{\text{roll}})$.
T, R	Homogeneous transformation matrix $\in SE(3)$ and rotation matrix $\in SO(3)$.
T^s, R^s, t^s	Transform, rotation, and translation matrices at arbitrary arc length s .
$\hat{R}^s$	Ground-truth rotation matrix measured from the motion capture system.
l	Vector of tendon lengths for a single section $([l_1, l_2, l_3]^T)$ or the full arm.
l_i	Length of the individual i -th tendon ($i \in \{1, 2, 3\}$).
l_j	Vector of tendon lengths for the j -th module ($j \in \{I, II, III\}$).
δ_i	Circumferential arrangement angle of the i -th tendon.
$\bar{l}$	Mean tendon actuation vector estimated from the workspace dataset.
δl	Tendon displacement command issued to the actuators.

all markers during data acquisition. To overcome this, we trained a mapping network $N_C : p \mapsto C$ to estimate ground-truth curvature parameters in the local reference frame of each module from tip pose, treating the robot as a modular extension of the single-section unit. Fig. S2b validates this approach against motion-capture ground truth, confirming it does not bias the evaluation pipeline as the error is much smaller than the one in Fig. S2a. However, N_C cannot replace vision-based sensing, as it is not distributed and would compound the inaccuracies of the pose prediction it relies on.

4.3 | Multi-Section Sampling and Bending Solver

Algorithm 1 details the statistically balanced sampling strategy [5] used to collect the multi-section dataset, relying on the

Bending Solver BS to enforce identical q -configurations across segments.

Algorithm 1 Multi-section sampling

```

Select randomly strategy  $s \in \{I, II, III\}$ 
if  $s = I$  then
    Sample  $l_I$ 
    Set  $l_{II}, l_{III} \leftarrow BS(l_I)$ 
else if  $s = II$  then
    Sample  $l_I, l_{II}$ 
     $l_{III} \leftarrow BS(l_{II})$ 
else
    Sample  $l_I, l_{II}, l_{III}$ 
end if

```

TABLE S2 | Notation and terminology: deep learning and control.

Symbol	Description
<i>Deep Learning Architectures</i>	
I	Stacked multi-view grayscale images from the embedded cameras.
n	Number of sections of the soft arm.
f_p^1, f_p^3	CNNs mapping images I to tip poses for the 1- and 3-section arms, respectively.
f_C^1, f_C^3	CNNs mapping images I to full-body curvature parameters for the 1- and 3-section arms.
N_l^1, N_l^3	MLPs for inverse kinematics (mapping targets to tendons) for the 1- and 3-section arms.
N_p^3	MLP for forward kinematics (3-section arm) mapping tendon lengths to tip poses.
N_C	MLP used to estimate ground-truth curvature parameters from tip poses during data collection.
$\hat{y}$	Network prediction output.
$\mathcal{L}$	Huber loss function used for network training.
η, ζ	Learning rate and Lagrange multiplier for secondary objective optimization during training.
$L_{\text{Grad-CAM}}$	Activation heatmap for spatial attention visualization.
A^k, α_k	Convolutional feature map channel and its computed importance weight for Grad-CAM.
<i>Control and Trajectory Planning</i>	
$p_{\text{goal}}, p_{\text{ref}}$	Final target pose and interpolated reference pose at the current time step.
$\mathcal{W}^1, \mathcal{W}^3$	Straight-line Cartesian path planner and $SE(3)$ SLERP trajectory planner.
ϵ, γ	Maximum waypoint spacing and convergence distance threshold.
$d_{SE(3)}, d_{SO(3)}$	Computed distance metrics for task-space poses and orientations.
β	Scaling coefficient linking rotational and translational distances in $SE(3)$.
$J, J^\#$	Analytical Jacobian matrix and its regularized Moore-Penrose pseudoinverse.
$\hat{J}_{t2l}, \hat{J}_{p2l}$	Estimated Jacobians derived from the data-driven mappings N_l^1 and N_l^3 .
λ	Damping parameter used for Jacobian regularization.
N	Null-space projection matrix $(I - J^\# J)$.
$H(l), \nabla H$	Secondary objective cost function and its projected gradient.
ξ_l	Arbitrary vector projected onto the null space for secondary task optimization.
BS	Bending Solver algorithm mapping actuation between sections during dataset sampling.

TABLE S3 | Single-section dataset fields.

Field	Description
I	Grayscale camera image
p	Tip pose
l	Tendon lengths
C	Curvature parameters (constant, affine, quadratic)

TABLE S4 | Multi-section dataset fields.

Field	Description
I	Three grayscale camera images
p_1, p_2	Intermediate tip poses
p	Final tip pose
l	Tendon lengths
C_1, C_2, C_3	Curvature parameters (constant, affine, quadratic)

The *Bending Solver* BS provides the linear mapping required to enforce the same q -configuration across segments. Given the circumferential arrangement of the tendons at angles $\delta = [\delta_1, \delta_2, \delta_3]^T$, we define the coefficient matrix $\mathbf{A}(\delta)$ as:

$$\mathbf{A}(\delta) = \begin{bmatrix} \cos(\delta_1) & \sin(\delta_1) & -1 \\ \cos(\delta_2) & \sin(\delta_2) & -1 \\ \cos(\delta_3) & \sin(\delta_3) & -1 \end{bmatrix} \quad (\text{S13})$$

To map the tendon lengths $\mathbf{l}_{old}$ from a source segment to a target segment with different pulley orientations δ_{new} , we compute:

$$\mathbf{l}_{new} = \mathbf{A}(\delta_{new}) \cdot (\mathbf{A}(\delta_{old})^{-1} \cdot \mathbf{l}_{old}) \quad (\text{S14})$$

This allows the system to translate the physical actuation of one module into an equivalent state for another.

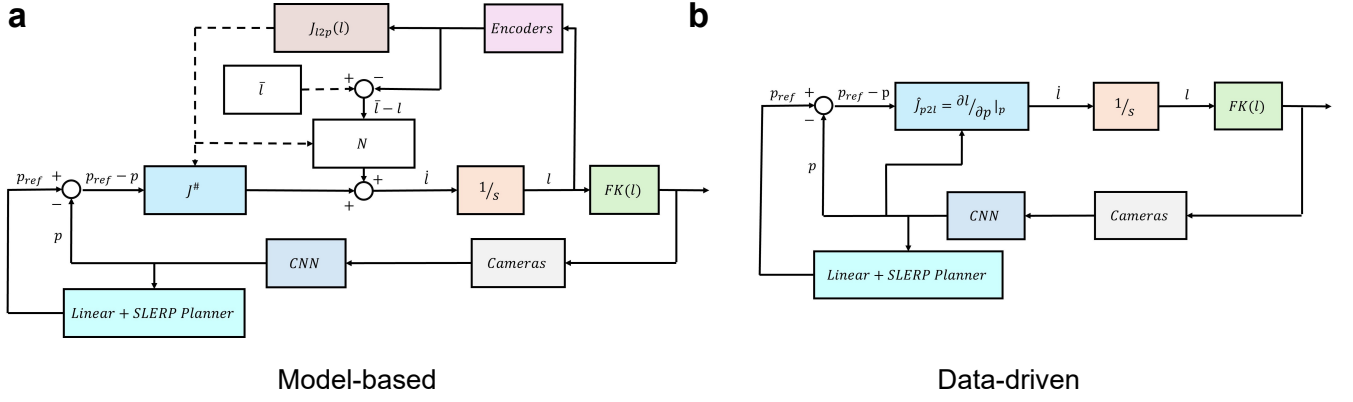


FIGURE S3 | (a) Model-based control scheme for the multi-section soft manipulator based on camera feedback. This approach uses model-based null-space optimization to translate observed poses into actuator commands. (b) Data-driven control scheme for the multi-section soft manipulator based on camera feedback. This approach uses data-driven kinematics learning to translate observed poses into actuator commands.

5 | Control Architectures

5.1 | Control Blocks Diagrams

The block diagrams in Fig. S3 illustrate the closed-loop architecture of the model-based and data-driven controllers for the multi-section arm, detailing the signal flow from camera feedback to actuator commands.

5.2 | Trajectory Planning and Distance Metrics

To generate trajectories between two poses and evaluate convergence, the multi-section controller relies on an $SE(3)$ planner $\mathcal{W}^3$ and a combined distance metric [6] over position and orientation. The angular distance in $SO(3)$ is defined as:

$$d_{SO(3)}(r_i, r_j) = \arccos \frac{\text{tr}(R_j R_i^T) - 1}{2} \quad (\text{S15})$$

and the combined $SE(3)$ distance is:

$$d_{SE(3)}(p_i, p_j) = \|t_i - t_j\|_2 + \beta d_{SO(3)}(r_i, r_j) \quad (\text{S16})$$

with $\beta = 0.2 \cdot \frac{180}{\pi}$ mm, so that a 5° error in orientation corresponds to 1 mm in position.

6 | Additional Results

6.1 | Absolute Pose Errors for the Single-Section Arm

Table S5 details the individual coordinate translation and rotation errors estimated by the f_p^1 CNN for the single-section arm.

6.2 | Relative Pose Errors for the Multi-Section Arm

Table S6 reports the relative pose errors of each section for the multi-section arm, computed with respect to the preceding segment. Module II contributes the most to the degradation in accuracy, both in terms of position and orientation.

TABLE S5 | Translation and rotation errors for the single-section arm.

Position [mm]		Orientation [°]	
x	2.28 ± 1.94	Yaw	0.42 ± 0.35
y	2.33 ± 2.02	Pitch	1.98 ± 1.67
z	1.22 ± 0.95	Roll	2.00 ± 1.71
Translation Error	3.99 ± 2.23	Rotation Error	3.11 ± 1.96

TABLE S6 | Relative pose errors for the multi-section arm.

Section	Position [mm]	Orientation [°]
I	4.44 ± 2.68	6.78 ± 15.94
II	11.92 ± 25.10	7.16 ± 16.02
III	8.86 ± 5.50	7.00 ± 13.17

6.3 | Control Workspace Definition

To distribute target points uniformly across the reachable workspace for controller evaluation, we apply an iterative k -medoids algorithm with $K = 20$. The same procedure is applied independently to the single-section and multi-section workspaces. Let $\mathcal{W} = \{x_1, \dots, x_{10,000}\} \subset \mathbb{R}^3$ denote the sampled workspace for a given robotic arm. The control workspace is defined as:

$$\mathcal{W}_{\text{control}} := \arg \min_{\{\mu_k \in \mathcal{W}\}_{k=1}^K} \sum_{k=1}^K \sum_{x_i \in \mathcal{C}_{\mu_k}} \|x_i - \mu_k\|_2^2 \quad (\text{S17})$$

where $\mathcal{C}_{\mu_k} = \{x_i \in \mathcal{W} : \|x_i - \mu_k\|_2 \leq \|x_i - \mu_j\|_2 \forall j\}$ is the cluster assigned to medoid μ_k .

6.4 | Single-Section Control Error Distribution

Fig. S4 shows the spatial distribution of closed-loop tracking errors on the control workspace for the single-section arm, for the model-based and data-driven controllers.

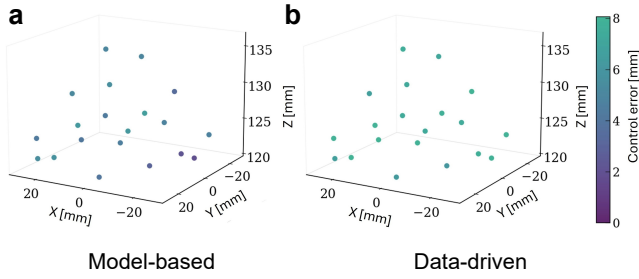


FIGURE S4 | Closed-loop tracking error distribution (mm) on the control workspace for the single-section arm. (a) Model-based. (b) Data-driven.

6.5 | Multi-Section Error Distribution

Fig. S5 illustrates the spatial distribution of closed-loop tracking errors, both in position and orientation, across the control workspace for the multi-section arm, comparing the model-based and data-driven approaches. Fig. S6 shows how this performance is affected by a disturbance, contrasting closed-loop and open-loop tracking under a 50 g tip payload.

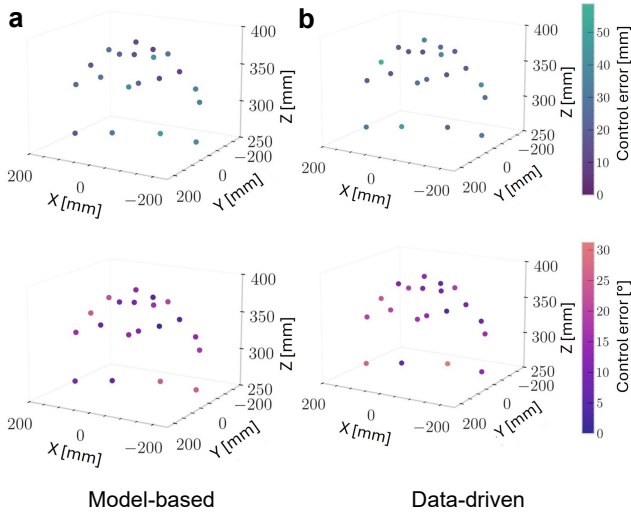


FIGURE S5 | Closed-loop tracking error distribution (mm and °) on the control workspace for the multi-section arm. (a) Model-based. (b) Data-driven.

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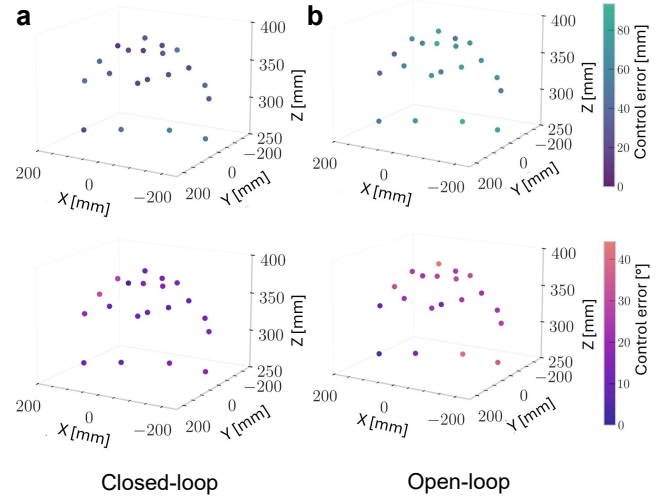


FIGURE S6 | Tracking error distribution (mm and °) for the multi-section arm under external disturbance. (a) Closed-loop. (b) Open-loop.

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